

MatRisk Lab

Classification
Internal – R&D

Repository
vikashg450/MatRisk-AI

Framework
Streamlit · PyTorch

Phase
4 – Technical Report

"Atoms → Physics → Finance. Material degradation is invisible until it isn't.
MatRisk AI makes it legible."



EXECUTIVE SUMMARY & ARCHITECTURE

01

The "Atoms to Dollars" Vision

Most infrastructure risk models skip foundational physical chemistry. MatRisk AI closes this gap by translating atomistic material properties into survival probabilities — and finally into dollar-denominated financial risk. The prototype operates as a clean, five-page Streamlit dashboard tracking the full lifecycle of physical assets.

MODULE A Material Lab Computes Bulk/Shear moduli and Poisson's Ratio using physics-informed neural networks with thermodynamic constraints.		MODULE B Risk Command Center Calculates asset-level Expected Loss (EL) and Exposure at Default (EAD) from material degradation signals.	
MODULE C Historical Validation Backtests survival models against real-world structural failure records across 5,000 bridge datasets.		MODULE D ESG & Market Signals Computes portfolio-wide carbon savings and generates trading alpha from material quality trends (MQI).	
PyTorch	PyCox / DeepSurv	Scikit-Learn	XGBoost

Matminer	Pandas	NumPy	Streamlit
----------	--------	-------	-----------

02

CORE AI FRAMEWORK

Physics-Informed Material Intelligence

The material intelligence layer uses a Physics-Informed Neural Network (PINN) to enforce thermodynamic validity throughout training. By penalizing violations of the fundamental elastic relationship, the network cannot produce physically impossible predictions.

TARGET PROPERTIES: K (BULK MODULUS), G (SHEAR MODULUS), ν (POISSON'S RATIO)

$$E = 2G(1 + \nu) = 3K(1 - 2\nu)$$

Stability bounds enforced: $K > 0 \mid G > 0 \mid -1 < \nu < 0.5$

Before physical testing, materials are subjected to **economic constraints**. By integrating element price histories (DS5), the system ensures that novel or recycled alloy compositions are financially and structurally viable.

3 TARGET PROPERTIES	0 THERMODYNAMIC VIOLATIONS	DS5 PRICE HISTORY SOURCE
------------------------	-------------------------------	-----------------------------

03

INFRASTRUCTURE RISK & SURVIVAL ANALYSIS

Predicting Remaining Useful Life

To translate material properties into structural longevity, MatRisk uses a **DeepSurv** (Cox Proportional Hazards) neural network trained on 5,000 bridge records (DS3). Key features include corrosion rate, age, design life, and tensile strength. An event (failure) is triggered when an asset becomes structurally deficient or its condition rating drops to 3 or below.

METRIC 01 / PD

Probability of Default

$PD(t) = 1 - S(t|x)$ Likelihood of structural failure at the current age, derived from the survival function.

METRIC 02 / EAD

Exposure at Default

Equal to the outstanding loan amount on the infrastructure asset. Captures the financial stake at risk.

CREDIT RISK TRANSLATION FORMULA

$$EL = PD \times LGD \times EAD$$

Loss Given Default assumed at 65% | EAD = outstanding infrastructure loan

"By unifying PINN material simulations with DeepSurv failure metrics, MatRisk closes the intelligence gap for material-dependent financial assets."

MatRisk Lab — Phase 4 Conclusion

VALIDATION & MARKET INTELLIGENCE

04

Historical Calibration & Alpha Generation

Survival models are validated against historical structural failures (DS6). The model demonstrates that higher predicted PD values accurately align with catastrophic and major real-world failures — confirming the predictive power of integrating corrosion and age into the DeepSurv framework.

SIGNAL / BUY

Bullish Signal

21-day MQI trend exceeds +2.0 → triggers Buy recommendation for Steel HRC and related commodities.

SIGNAL / SELL

Bearish Signal

21-day MQI trend falls below -2.0 → triggers Sell/Short signal for exposed commodity positions.

Environmental Quantification

The ESG Calculator quantifies the environmental impact of switching from virgin to recycled materials, computing portfolio-wide emission reductions using per-asset carbon intensity data.

CARBON INTENSITY COMPARISON

Steel: Virgin vs. Recycled

Virgin Steel at 1.85 kg CO₂/kg vs. Recycled Steel at 0.40 kg CO₂/kg — a 78% reduction in embodied carbon per kilogram.

The PINN verifies that a simulated 5% drop in moduli — representing recycled material degradation — still meets required structural safety limits, ensuring ESG-motivated material substitution does not compromise asset integrity.

IMPLEMENTATION METRICS

Key Technical Details

METRIC	DATA SOURCE	IMPLEMENTATION DETAIL
RUL Validation	DS6 — Historical Failures	Catastrophic failures correspond accurately to high predicted PD values.
Elastic Consistency (PINN)	DS1 — Material Properties	$E = 2G(1+\nu) = 3K(1-2\nu)$ enforced via PyTorch custom loss function.
Carbon Savings (ESG)	DS3 — Bridge Infrastructure	Computed per asset using deck area and virgin vs. recycled carbon intensity factors.
Alpha Signals (MQI)	Market Data	21-day momentum threshold of ± 2.0 triggers directional commodity trading signals.